

Sales & Customer Performance Dashboard Project

Project Overview

This project focuses on cleaning, analyzing, and visualizing a sales and customer performance dataset using Microsoft Excel and Power BI. The objective was to transform raw business data into an interactive dashboard that provides meaningful insights into sales performance, profitability, customer satisfaction, and shipping outcomes to support data-driven decision-making.

Objectives

The main objectives of this project were to:

- **Clean and prepare the raw dataset for analysis.**
- **Summarize key business metrics using Pivot Tables.**
- **Build an interactive Power BI dashboard.**
- **Identify sales trends, profitable regions, and customer performance.**
- **Present insights in a clear and visually appealing format.**

Data Cleaning Process

Before analysis, the dataset was cleaned in Microsoft Excel to improve data quality and accuracy. The cleaning process included:

- **Removing duplicate records.**
- **Handling missing or incomplete values.**
- **Correcting inconsistent text formatting.**
- **Standardizing data formats such as dates and numbers.**
- **Renaming columns for clarity.**
- **Creating calculated fields where necessary.**
- **Verifying data accuracy before importing it into Power BI.**

Data Analysis

After cleaning the data, Pivot Tables were used to summarize and explore the dataset. The analysis focused on:

- **Total Sales**
- **Total Profit**
- **Total Orders**
- **Average Customer Satisfaction**
- **Sales by Product Category**
- **Top 10 Cities by Profit**
- **Monthly Profit by Region**
- **Impact of Shipping Delays on Product Returns**

These summaries helped identify trends and informed the design of the dashboard.

Dashboard Features

The Power BI dashboard includes:

- **KPI cards displaying Total Sales, Total Profit, Total Orders, and Average Customer Satisfaction.**
- **Treemap showing the Top 10 Cities with the highest profit.**
- **Bar chart comparing Sales by Product Category.**
- **Line chart illustrating Monthly Profit by Region.**
- **Pie chart showing the relationship between Shipping Delays and Product Returns.**

Key Insights

The dashboard revealed several important business insights:

- **Electronics generated the highest sales among all product categories.**
- **Kolkata recorded the highest profit compared to other cities.**
- **Profit trends varied across different regions throughout the year.**
- **Shipping delays had a noticeable impact on product returns.**
- **The average customer satisfaction score of 2.81 indicates opportunities to improve customer experience.**

Tools Used

- **Microsoft Excel (Data Cleaning and Pivot Tables)**
- **Power BI (Dashboard Development and Data Visualization)**

Skills Demonstrated

This project demonstrates my ability to:

- **Clean and prepare raw datasets.**
- **Perform exploratory data analysis.**
- **Create Pivot Tables for business reporting.**
- **Build interactive Power BI dashboards.**
- **Develop KPI metrics and calculated measures.**
- **Design professional business reports.**
- **Communicate insights through data visualization.**

Conclusion

This project demonstrates the complete data analysis workflow, from data cleaning and preparation to visualization and business insight generation. It highlights my ability to transform raw data into actionable information that supports informed business decisions and showcases my proficiency in Excel and Power BI.